Architecture: implemented lanes

Current components and explicit target boundaries | 26 October 2026

LOCAL INGESTION

PDF / OCR
Scans need a text layer

→

GROBID
Structured TEI XML

→

LightRAG
Local graph + vectors

→

Neon dev import
Frozen batch / pgvector

Current graph: 5,101 nodes / 6,240 edges. Imported snapshot: 3 documents / 7,582 embeddings.

QUESTION AND ANSWER

Oracle UI
Question + user context

→

Python server
Render / local dev

→

Evidence retrieval
Neon FTS; dev graph adapter

→

Claude
Evidence to cited answer

Response streams back through the server to Oracle.

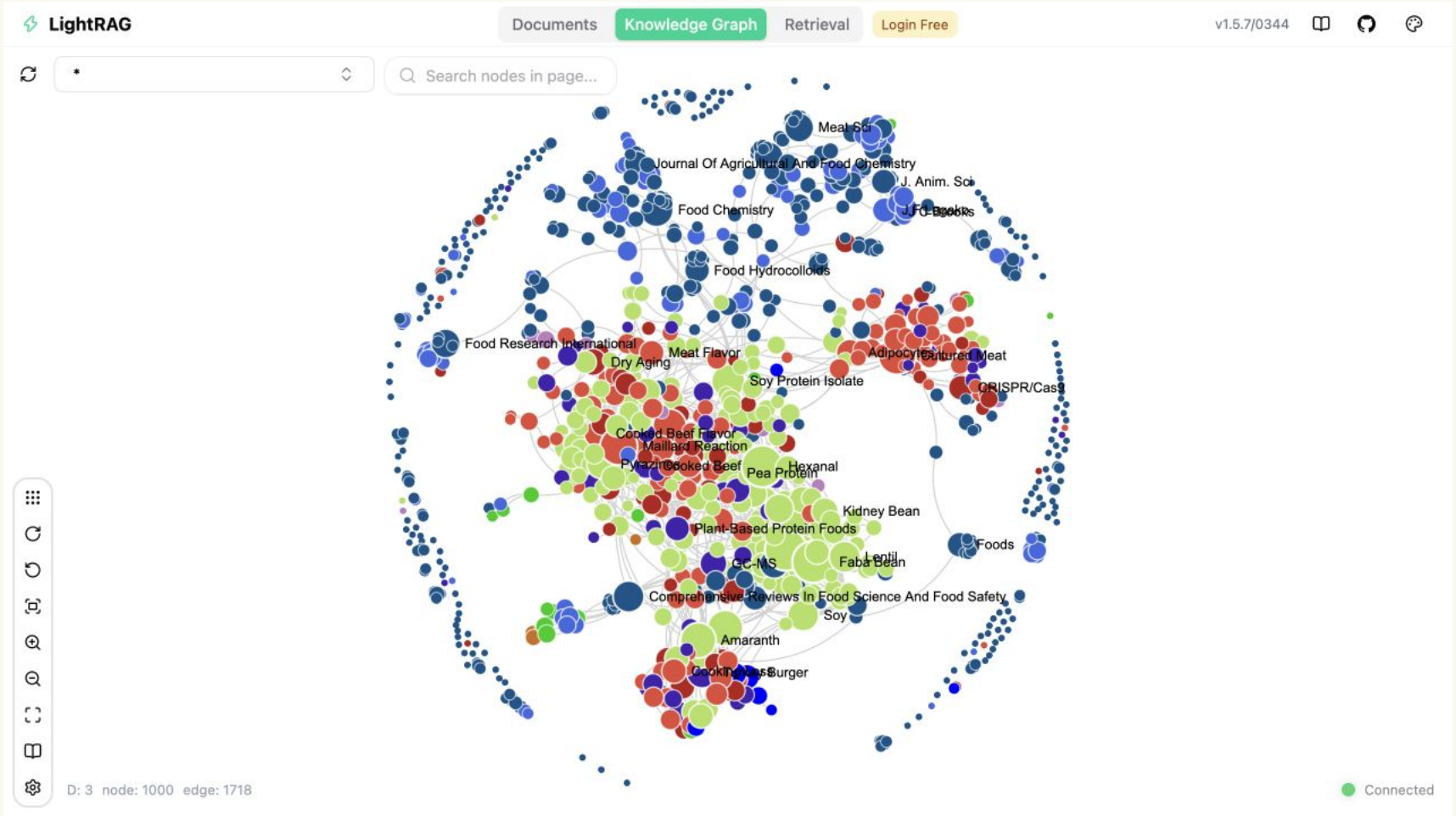
Selected target: independent M2 curation before admission

Qwen extraction / M2 validation / GPT reasoning are target roles. Current runtime uses Claude for Oracle; the full target is not implemented.

Production version verified; diagram does not assert an end-to-end model run or deployed graph retrieval.

LightRAG: the extracted graph

Actual web interface capture | Overview subset, not the full graph



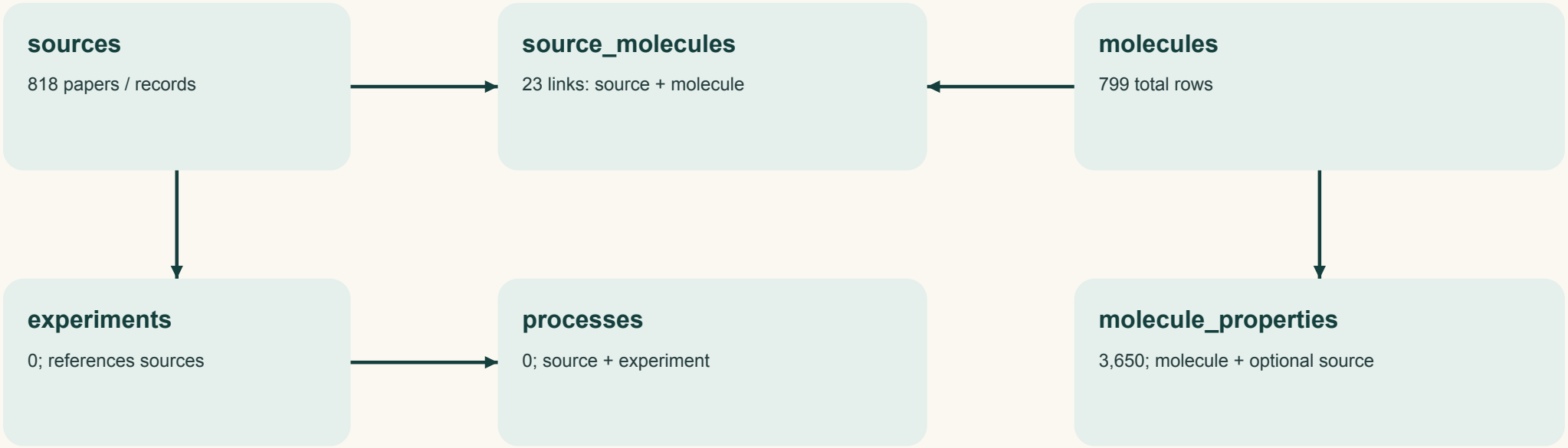
Visible view: 1,000 nodes / 1,718 edges, depth 3. Full stored GraphML: 5,101 nodes / 6,240 edges.

This is an extracted knowledge graph. Node counts and connections do not establish scientific correctness or source agreement.

Source: local LightRAG v1.5.7/0344; captured 26 October 2026. No graph edits or model queries.

Schema: where evidence connects

Simplified map of inspected foreign keys | Full inventory is in schema/SCHEMA_REFERENCE.md



OTHER CANONICAL DOMAINS

claims (45), experts (3,129), organizations, reactions, sensory descriptors, taxonomy and audit tables. Not every domain is connected by an implemented FK; the full inventory is authoritative.

Dev-only lightrag_demo schema

batches > documents > chunks; entities > relations; embeddings > evidence > chunks. Documents reference public.sources. 7 added tables; vector(1024), HNSW cosine and FTS indexes.

Production: 49 base tables + 16 views. Dev: 56 base tables + 16 views. Arrows show reference relationships, not execution.